

Fluent SQL Query Architect

Modern Object-Relational Mapping (ORM) frameworks utilize "Fluent Interfaces" to enable the incremental construction of complex database queries. By leveraging method chaining, developers can create readable and maintainable SQL statements while ensuring structural integrity through deferred execution.

Implement a `QueryBuilder` class that constructs standardized SQL SELECT statements using a chained API. The architect must handle column selection, table specification, and logical filtering (WHERE clauses), while enforcing a "Fail-Fast" policy that validates mandatory components (like table names) before final string generation.

Example 1 Input:

```
QUERY SELECT name age FROM users WHERE age > 21 BUILD
```

Output:

```
SELECT name, age FROM users WHERE age > 21
```

Explanation: The builder chains `.select()`, `.from_table()`, and `.where()` methods. The final `.build()` call joins these components into a valid SQL string.

Example 2 Input:

```
QUERY FROM products WHERE price < 100 BUILD
```

Output:

```
SELECT * FROM products WHERE price < 100
```

Explanation: If no columns are explicitly selected, the builder defaults to the `*` wildcard for the SELECT clause.

Function Parameter:

- `columns (tuple)`: Variable number of column identifiers.

- `table_name` (str): The target database relation.
- `value` (Any): The comparison value (must be wrapped in single quotes if it is a string).

Returns

- `str`: The fully constructed SQL statement.

Constraints

- All builder methods (except `build`) must return `self` to facilitate chaining.
- Multiple `where` calls must be concatenated using the `AND` operator.
- Raise `ValueError("Table name not specified")` if `build` is called without a source table.

Input Format for Custom Testing Input from stdin will be processed as follows: A sequence of keywords: `SELECT`, `FROM`, `WHERE`, and `BUILD`, each followed by their respective parameters.

Sample Case 0 Sample Input

```
QUERY SELECT id FROM orders BUILD
```

Sample Output

```
SELECT id FROM orders
```

Explanation A simple query selecting only the ID column from the orders table.